HBase Assignment

Beginner & Intermediate

Individual Practice Task

Overview

In this assignment you will build and query a product inventory system using HBase. You will work with a table called store that holds information about products across two column families.

Table Name	Column Families	Total Records
store	details, inventory	5 products

Dataset — store Table

Column Family	Columns
details	name, brand, category
inventory	price, quantity, warehouse

Row Key	Name	Brand	Category	Price	Quantity	Warehouse
1	laptop	dell	electronics	1200	50	A
2	phone	samsung	electronics	800	120	A
3	desk	ikea	furniture	300	30	B
4	chair	ikea	furniture	150	60	B
5	tablet	apple	electronics	600	80	A

Instructions

1. Read each question carefully before writing your command.
2. Write your HBase shell command(s) in the answer box below each question.
3. Test every command in your HBase shell and verify the output before submitting.
4. The dataset above is your only reference — do not use the emp or student tables.

⚠ *Answers are provided at the end of this document. Do not look ahead until you have attempted all questions.*

PART A — BEGINNER

Questions 1–6 cover table creation, inserting data, scanning, getting, and filtering.

Q1 Create the store table with two column families: details and inventory.

Your Answer:

```
create 'store', 'details', 'inventory'
```

Q2 Insert all 5 rows from the dataset above into the store table. Insert all columns for both column families.

Your Answer:

```
 # Row 1: Laptop
put 'store', '1', 'details:name', 'laptop'
put 'store', '1', 'details:brand', 'dell'
put 'store', '1', 'details:category', 'electronics'
put 'store', '1', 'inventory:price', '1200'
put 'store', '1', 'inventory:quantity', '50'
put 'store', '1', 'inventory:warehouse', 'A'

# Row 2: Phone
put 'store', '2', 'details:name', 'phone'
put 'store', '2', 'details:brand', 'samsung'
put 'store', '2', 'details:category', 'electronics'
put 'store', '2', 'inventory:price', '800'
put 'store', '2', 'inventory:quantity', '120'
put 'store', '2', 'inventory:warehouse', 'A'

# Row 3: Desk
put 'store', '3', 'details:name', 'desk'
put 'store', '3', 'details:brand', 'ikea'
put 'store', '3', 'details:category', 'furniture'
put 'store', '3', 'inventory:price', '300'
put 'store', '3', 'inventory:quantity', '30'
put 'store', '3', 'inventory:warehouse', 'B'

# Row 4: Chair
put 'store', '4', 'details:name', 'chair'
put 'store', '4', 'details:brand', 'ikea'
put 'store', '4', 'details:category', 'furniture'
put 'store', '4', 'inventory:price', '150'
put 'store', '4', 'inventory:quantity', '60'
put 'store', '4', 'inventory:warehouse', 'B'

# Row 5: Tablet
put 'store', '5', 'details:name', 'tablet'
put 'store', '5', 'details:brand', 'apple'
put 'store', '5', 'details:category', 'electronics'
put 'store', '5', 'inventory:price', '600'
put 'store', '5', 'inventory:quantity', '80'
put 'store', '5', 'inventory:warehouse', 'A'
```

Q3 Scan the entire store table and confirm all 5 rows are present.

Your Answer:

```
 hbase(main):043:0> scan 'store'
ROW                     COLUMN+CELL
 1                      column=details:brand, timestamp=2026-05-24T12:32:20.835, v
                        alue=dell
 1                      column=details:category, timestamp=2026-05-24T12:32:20.862
                        , value=electronics
 1                      column=details:name, timestamp=2026-05-24T12:32:20.797, va
```

```
                      lue=laptop
 1                    column=inventory:price, timestamp=2026-05-24T12:32:20.894,
                       value=1200
 1                    column=inventory:quantity, timestamp=2026-05-24T12:32:20.9
                      12, value=50
 1                    column=inventory:warehouse, timestamp=2026-05-24T12:32:20.
                      933, value=A
 2                    column=details:brand, timestamp=2026-05-24T12:32:20.982, v
                      alue=samsung
 2                    column=details:category, timestamp=2026-05-24T12:32:20.997
                      , value=electronics
 2                    column=details:name, timestamp=2026-05-24T12:32:20.963, va
                      lue=phone
 2                    column=inventory:price, timestamp=2026-05-24T12:32:21.014,
                       value=800
 2                    column=inventory:quantity, timestamp=2026-05-24T12:32:21.0
                      28, value=120
 2                    column=inventory:warehouse, timestamp=2026-05-24T12:32:21.
                      051, value=A
 3                    column=details:brand, timestamp=2026-05-24T12:32:21.122, v
                      alue=ikea
 3                    column=details:category, timestamp=2026-05-24T12:32:21.136
                      , value=furniture
 3                    column=details:name, timestamp=2026-05-24T12:32:21.108, va
                      lue=desk
 3                    column=inventory:price, timestamp=2026-05-24T12:32:21.149,
                       value=300
 3                    column=inventory:quantity, timestamp=2026-05-24T12:32:21.1
                      65, value=30
 3                    column=inventory:warehouse, timestamp=2026-05-24T12:32:21.
                      179, value=B
 4                    column=details:brand, timestamp=2026-05-24T12:32:21.218, v
                      alue=ikea
 4                    column=details:category, timestamp=2026-05-24T12:32:21.229
                      , value=furniture
 4                    column=details:name, timestamp=2026-05-24T12:32:21.204, va
                      lue=chair
 4                    column=inventory:price, timestamp=2026-05-24T12:32:21.239,
                       value=150
 4                    column=inventory:quantity, timestamp=2026-05-24T12:32:21.2
                      51, value=60
 4                    column=inventory:warehouse, timestamp=2026-05-24T12:32:21.
                      262, value=B
 5                    column=details:brand, timestamp=2026-05-24T12:32:21.293, v
                      alue=apple
 5                    column=details:category, timestamp=2026-05-24T12:32:21.305
                      , value=electronics
 5                    column=details:name, timestamp=2026-05-24T12:32:21.283, va
                      lue=tablet
 5                    column=inventory:price, timestamp=2026-05-24T12:32:21.317,
                       value=600
 5                    column=inventory:quantity, timestamp=2026-05-24T12:32:21.3
                      29, value=80
 5                    column=inventory:warehouse, timestamp=2026-05-24T12:32:22.
                      924, value=A
5 row(s)
Took 0.0646 seconds
```

Q4 Retrieve all columns for the product with row key 3 (desk).

Your Answer:

```
hbase(main):001:0> get 'store', '3'

COLUMN                    CELL
 details:brand            timestamp=2026-05-24T12:32:21.122, value=ikea
 details:category         timestamp=2026-05-24T12:32:21.136, value=furniture
 details:name             timestamp=2026-05-24T12:32:21.108, value=desk
 inventory:price          timestamp=2026-05-24T12:32:21.149, value=300
 inventory:quantity       timestamp=2026-05-24T12:32:21.165, value=30
 inventory:warehouse      timestamp=2026-05-24T12:32:21.179, value=B

1 row(s)
Took 0.4203 seconds
```

Q5 Retrieve only the inventory:price column for the product with row key 5 (tablet).

Your Answer:

```
hbase(main):039:0> get 'store', '5', 'inventory:price'
COLUMN                      CELL
inventory:price           timestamp=2026-05-24T12:32:21.317, value=600
1 row(s)
Took 0.1081 seconds
```

Q6 Use a ValueFilter scan to find all products with warehouse value A.

Hint: Use the COLUMNS and FILTER options together in your scan.

Your Answer:

```
hbase(main):001:0> scan 'store', {COLUMNS => 'inventory:warehouse', FILTER =>
"ValueFilter(=, 'binary:A')"}

ROW                      COLUMN+CELL
 1                       column=inventory:warehouse, timestamp=2026-05-24T12:32:2
                         0.933, value=A
 2                       column=inventory:warehouse, timestamp=2026-05-24T12:32:2
                         1.051, value=A
 5                       column=inventory:warehouse, timestamp=2026-05-24T12:32:2
                         2.924, value=A

3 row(s)
Took 0.3891 seconds
```

PART B — INTERMEDIATE

Questions 7–12 cover versioning, advanced filters, altering the table, and managing table lifecycle.

Q7 Enable versioning on the inventory column family to store up to 4 versions.

Your Answer:

```
alter 'store', NAME => 'inventory', VERSIONS => 4
Updating all regions with the new schema...
1/1 regions updated.
Done.
Took 2.6093 seconds
```

Q8 Update the inventory:quantity for the laptop (row 1) three times: first to 45, then to 40, then to 35. Then retrieve all stored versions of that cell.

Hint: You need to alter versions first — make sure Q7 is done before this.

Your Answer:

```
 hbase(main):001:0> put 'store', '1', 'inventory:quantity', '45'
Took 0.4150 seconds
hbase(main):002:0> put 'store', '1', 'inventory:quantity', '40'
Took 0.0032 seconds
hbase(main):003:0> put 'store', '1', 'inventory:quantity', '35'
Took 0.0032 seconds
hbase(main):004:0> get 'store', '1', {COLUMN => 'inventory:quantity', VERSIONS => 4}
COLUMN                CELL
 inventory:quantity   timestamp=2026-05-24T12:46:45.002, value=35
 inventory:quantity   timestamp=2026-05-24T12:46:43.690, value=40
 inventory:quantity   timestamp=2026-05-24T12:46:43.657, value=45
 inventory:quantity   timestamp=2026-05-24T12:32:20.912, value=50
1 row(s)
Took 0.0877 seconds
```

Q9 Use SingleColumnValueFilter to find all products in the furniture category.

Your Answer:

```
 hbase(main):001:0> scan 'store', {FILTER => "SingleColumnValueFilter('details',
'category', =, 'binary:furniture')"}
ROW                   COLUMN+CELL
 3                    column=details:brand, timestamp=2026-05-24T12:32:21.122, v
                      alue=ikea
 3                    column=details:category, timestamp=2026-05-24T12:32:21.136
                      , value=furniture
 3                    column=details:name, timestamp=2026-05-24T12:32:21.108, va
                      lue=desk
 3                    column=inventory:price, timestamp=2026-05-24T12:32:21.149,
                       value=300
 3                    column=inventory:quantity, timestamp=2026-05-24T12:32:21.1
                      65, value=30
 3                    column=inventory:warehouse, timestamp=2026-05-24T12:32:21.
                      179, value=B
 4                    column=details:brand, timestamp=2026-05-24T12:32:21.218, v
                      alue=ikea
 4                    column=details:category, timestamp=2026-05-24T12:32:21.229
                      , value=furniture
 4                    column=details:name, timestamp=2026-05-24T12:32:21.204, va
```

```
                      lue=chair
 4                    column=inventory:price, timestamp=2026-05-24T12:32:21.239,
                       value=150
 4                    column=inventory:quantity, timestamp=2026-05-24T12:32:21.2
                      51, value=60
 4                    column=inventory:warehouse, timestamp=2026-05-24T12:32:21.
                      262, value=B
2 row(s)
Took 0.4212 seconds
```

Q10 Use SingleColumnValueFilter to find all products stored in warehouse B.

Your Answer:

```
 hbase(main):001:0> scan 'store', {FILTER => "SingleColumnValueFilter('inventory',
'warehouse', =, 'binary:B')"}
ROW                   COLUMN+CELL
 3                    column=details:brand, timestamp=2026-05-24T12:32:21.122, v
                      alue=ikea
 3                    column=details:category, timestamp=2026-05-24T12:32:21.136
                      , value=furniture
 3                    column=details:name, timestamp=2026-05-24T12:32:21.108, va
                      lue=desk
 3                    column=inventory:price, timestamp=2026-05-24T12:32:21.149,
                       value=300
 3                    column=inventory:quantity, timestamp=2026-05-24T12:32:21.1
                      65, value=30
 3                    column=inventory:warehouse, timestamp=2026-05-24T12:32:21.
                      179, value=B
 4                    column=details:brand, timestamp=2026-05-24T12:32:21.218, v
                      alue=ikea
 4                    column=details:category, timestamp=2026-05-24T12:32:21.229
                      , value=furniture
 4                    column=details:name, timestamp=2026-05-24T12:32:21.204, va
                      lue=chair
 4                    column=inventory:price, timestamp=2026-05-24T12:32:21.239,
                       value=150
 4                    column=inventory:quantity, timestamp=2026-05-24T12:32:21.2
                      51, value=60
 4                    column=inventory:warehouse, timestamp=2026-05-24T12:32:21.
                      262, value=B
2 row(s)
Took 0.3968 seconds
```

Q11 Add a new column family called supplier to the store table. Then insert supplier:country as egypt for row 1 and supplier:country as korea for row 2. Verify the schema updated correctly.

Your Answer:

```
 #Step 1: Add the New Column Family

hbase(main):001:0> alter 'store', NAME => 'supplier'
Updating all regions with the new schema...
1/1 regions updated.
Done.
Took 3.0380 seconds

Step 2: Insert the Supplier Data
hbase(main):001:0> put 'store', '1', 'supplier:country', 'egypt'
Took 0.4211 seconds
hbase(main):002:0> put 'store', '2', 'supplier:country', 'korea'
Took 0.0033 seconds

Step 3: Verify the Schema

hbase(main):001:0> describe 'store'
Table store is ENABLED
store
COLUMN FAMILIES DESCRIPTION
{NAME => 'details', BLOOMFILTER => 'ROW', IN_MEMORY => 'false', VERSIONS => '1',
 KEEP_DELETED_CELLS => 'FALSE', DATA_BLOCK_ENCODING => 'NONE', COMPRESSION => 'N
ONE', TTL => 'FOREVER', MIN_VERSIONS => '0', BLOCKCACHE => 'true', BLOCKSIZE =>
'65536', REPLICATION_SCOPE => '0'}

{NAME => 'inventory', BLOOMFILTER => 'ROW', IN_MEMORY => 'false', VERSIONS => '4
', KEEP_DELETED_CELLS => 'FALSE', DATA_BLOCK_ENCODING => 'NONE', COMPRESSION =>
'NONE', TTL => 'FOREVER', MIN_VERSIONS => '0', BLOCKCACHE => 'true', BLOCKSIZE =
> '65536', REPLICATION_SCOPE => '0'}

{NAME => 'supplier', BLOOMFILTER => 'ROW', IN_MEMORY => 'false', VERSIONS => '1'
, KEEP_DELETED_CELLS => 'FALSE', DATA_BLOCK_ENCODING => 'NONE', COMPRESSION => '
NONE', TTL => 'FOREVER', MIN_VERSIONS => '0', BLOCKCACHE => 'true', BLOCKSIZE =>
 '65536', REPLICATION_SCOPE => '0'}

3 row(s)
```

Q12 CHALLENGE 🏆 Write a single scan that returns only products that are in the electronics category AND stored in warehouse A.

💡 *Hint: Look up FilterList with MUST_PASS_ALL to combine two SingleColumnValueFilter conditions.*

Your Answer:

```
 scan 'store', {FILTER => "SingleColumnValueFilter('details', 'category', =,
'binary:electronics') AND SingleColumnValueFilter('inventory', 'warehouse', =, 'binary:A')"}
ROW                      COLUMN+CELL
 1                       column=details:brand, timestamp=2026-05-24T12:32:20.835, v
                         alue=dell
 1                       column=details:category, timestamp=2026-05-24T12:32:20.862
                         , value=electronics
 1                       column=details:name, timestamp=2026-05-24T12:32:20.797, va
                         lue=laptop
 1                       column=inventory:price, timestamp=2026-05-24T12:32:20.894,
                          value=1200
 1                       column=inventory:quantity, timestamp=2026-05-24T12:46:45.0
                         02, value=35
 1                       column=inventory:warehouse, timestamp=2026-05-24T12:32:20.
                         933, value=A
 1                       column=supplier:country, timestamp=2026-05-24T12:54:44.051
                         , value=egypt
 2                       column=details:brand, timestamp=2026-05-24T12:32:20.982, v
                         alue=samsung
 2                       column=details:category, timestamp=2026-05-24T12:32:20.997
                         , value=electronics
 2                       column=details:name, timestamp=2026-05-24T12:32:20.963, va
                         lue=phone
 2                       column=inventory:price, timestamp=2026-05-24T12:32:21.014,
                          value=800
 2                       column=inventory:quantity, timestamp=2026-05-24T12:32:21.0
                         28, value=120
 2                       column=inventory:warehouse, timestamp=2026-05-24T12:32:21.
                         051, value=A
 2                       column=supplier:country, timestamp=2026-05-24T12:54:45.692
                         , value=korea
 5                       column=details:brand, timestamp=2026-05-24T12:32:21.293, v
                         alue=apple
 5                       column=details:category, timestamp=2026-05-24T12:32:21.305
                         , value=electronics
 5                       column=details:name, timestamp=2026-05-24T12:32:21.283, va
                         lue=tablet
 5                       column=inventory:price, timestamp=2026-05-24T12:32:21.317,
                          value=600
 5                       column=inventory:quantity, timestamp=2026-05-24T12:32:21.3
                         29, value=80
 5                       column=inventory:warehouse, timestamp=2026-05-24T12:32:22.
                         924, value=A
3 row(s)
Took 0.4303 seconds
```